

SHEKHAR BHATTARAI

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📍 Toulouse, France

EDUCATION

M.Sc. Satellite Communication System (SATCOM)

[2023-2025(ongoing)]

INP TOULOUSE, ENSEEIHT | Toulouse, France

Grades : M1(13.628 / 20)

B.Eng. in Electronics and Communication Engineering

[2017-2022]

Tribhuvan University, Institute of Engineering, Eastern Regional Campus | Dharan, Nepal

Final Percentage: 65.54 %

- **Relevant Courses:** Embedded system, Electromagnetics, RF and Microwave, antenna and wave propagation, Wireless communication, Satellite Com. system, & Remote Sensing, Broadcast

BACHELOR THESIS

Major project: "Design and Performance Analysis of Different Microstrip Patch Antenna for 5G Application"

- Design of MPA at 3.2 GHz using different substrate to determine the effect of substrate on its Gain and radiation efficiency.

TECHNICAL SKILLS

RF and Microwave: EM theory (Coupling, S-parameters), Active/Passive RF, Antenna theory, Analog/Digital Beamforming algorithms, MIMO, Wave Propagation

Wireless and SATCOM: NTN Standards (Rel. 17/18), Probability theory, Modulation techniques, Channel coding, multiple access, diversity techniques, Software define Radio (SDR), Digital and Statistical Signal Processing, Radar applications

Programming: MATLAB/Simulink (Antenna, Satcom, 5G, NTN toolbox), Python, C/C++

Simulation & Analysis: Keysight ADS, ANSYS HFSS, CST, Altair FEKO, COMSOL Multiphysics, GNU radio

Embedded Systems & Hardware: SDRs (USRP, HackRF), FPGA (VHDL), Embedded design for RF applications, RF measurement tools, STM and other families of microcontrollers.

Other: Linux, Git/version control, MS Office, SolidWorks

EXPERIENCES

Intern as a Satellite communications system Engineer

[Feb. 2025 - Aug.2025(ongoing)]

Continental Automotive France S.A.S, 31036 Toulouse, France

- As an end of the study intern, I am working on review on latest 3GPP/5GAA standards, link budget analysis of different LEO constellations for automotive application (sub 6 GHz and Ka/Ku band), Parametric study, modeling and validation using digital twin models to recommend key parameters for next gen. NTN vehicle terminals.

Intern as a R&D on compact phased array antenna

[June 2024- Sep 2024]

Anywaves, 31000 Toulouse, France

- Design validation, simulation and analysis of phased array antenna on surface waves, mutual coupling on scan angle, improving active S-parameters to enhance realized gain and to avoid scan blindness on X-band satellite payload antenna.
- Integrated radome structures to enhance beam steering, realized gain, cross-polarization discrimination (XPD), and sidelobe suppression, achieving target performance metrics.

Research and Development Engineer

[Feb 2022 – Jul 2023]

Orion space Nepal pvt. Ltd., Nepal

- Antenna design, tuning and installation for Pico satellites and Ground Station receiver.
- R&D on Embedded and Software development for commercial PocketQubes (nano/Pico satellites).
- Testing and implementation of Several modulation techniques: CW, GFSK and LoRa for Satellite communication in UHF band.
- Link budget design, power budget, Resource allocation and management for Pico satellites.

PROJECTS

RF link budget design for the Super Heterodyne Tx Architecture

- Budget analysis of RF payload of a telecom transmitter/receiver at 900 MHz/4 GHz, and design using COTs available to meet target requirement, where I took the responsibility to design HPA at 4 GHz.

Design of dual band bandpass filter and phase shifter using Capacitive plate MEMS

- Design of capacitive MEMS-based dual bandpass filter with center frequency 60 GHz & 40 GHz.
- A dynamic phase shifter was also designed using the same MEMS.

Design of Enhanced Gain Power Amplifier at 600 MHz

- Design of an amplifier with optimized gain was studied. Complementary analysis to characterize the efficiency (PAE) and the linearity of the amplifier (IP3, C/I3), power limitations (P1dB), gain and noise circles using ADS.

Telecom spacecraft sizing for GEO (AIRBUS Space & Defense)

- RF Link budget, antenna selection and placement, power budget, thermal sizing and heat dissipation, LEOP Propellant, mass budget and hardware trade-off and optimization to comply with launcher and mission constraints.

Design of finite planner array antenna at 2.5 GHz

- Design of unit element of MPA at 2.5 GHz, optimizing using Quarter wavelength inverter matching and studied infinite arrays, ideal finite arrays and finite real arrays by performing full wave simulation.

Design and Simulation of Horn and Patch Antennas

- Design of X-band Horn antenna with gain of 18dBi.
- Design of patch antenna using Rogers RO4003 and analyze the different feeding techniques and Complexity in design.

ADDITIONAL COURSES

- **KARI international Space Training (KARIST) 2022**, Korean Aerospace Research Institute (2022)
- **RF and Millimeter-wave Circuit Design**, Eindhoven University of Technology, Coursera (2024)

LANGUAGE SPOKEN

- Nepali (Native)
- English (fluent)
- Hindi (fluent)
- French (A2)

PRESENTATION

- **Keynote Speaker on Pico-Sat for the space Pioneer**
Yuri's Night Celebration 2023, St. Xavier's college Nepal | 2023
- **Present Scenario of Satellite Development in Nepal**
IOE Pulchowk campus, Pulchowk Nepal | 2023

HONORS & AWARDS

- **Participant, International Agriculture Robotics Competition.**
United Nation ESCAP-CSAM, Qingdao, China 2019
- **Runner- Up IOT Smart Digital Room design**
X-Tech Studio 1.0 Exposition, IOE Eastern Regional Campus, 2018

LEADERSHIP & VOLUNTERING

- Technical coordinator, DELTA 2.0 (National Level Engineering Event)
- Trainer Basic Robotics and Microcontrollers, Robotics Club and EXCESS
- Trainer in educational CanSat and CubeSat Technologies
- Member, Robotics Club Eastern Regional Campus Dharan, (2017-2022)

REFERENCES

Marie-Laure BOUCHERET

IRIT/ENSEEIH (Program Coordinator SATCOM)

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Hamza KAOUACH

LAPLACE/ assoc Prof. Toulouse INP ENESSIHT

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